

Financial Scars: Prior Financial Distress and Risk-Taking Behavior among U.S. Households

University of Oklahoma
Data Science for Economists

Todvwa Dlamini

May 14, 2026

1 Introduction

Current economic conditions, in addition to shaping household financial decisions, influence previous financial experiences. The general economic theory is based on the assumption that people with given preferences maximize expected utility when deciding on their finances. The crucial feature of this framework is that past financial experiences have only an indirect impact through their impact on wealth and income. Behavioral economics theory, however, indicates that previous experiences can have a separate effect on the risk and uncertainty perception, which can subsequently influence future decision-making.

Prospect Theory [1] is one important framework that has been developed to address this process. The theory suggests that people think in terms of outcomes in relation to a reference point, and losses have more impact on decisions than gains of the same value. Previous studies have focused on framed decisions within explicit gambles, but there is less understanding of whether financial ill health has long-lasting behavioral consequences when decisions are made in real life.

This paper investigates if households who experienced financial distress in the past were more likely to be conservative in their financial actions over the subsequent year. The study does not directly test loss aversion, but rather whether financial distress households decrease their exposure to risky financial assets. This distinction is crucial as it means that household portfolio decisions reflect financial behavior rather than actual experimental choices in the face of risk.

I create a measure of financial distress for the individuals based on negative net worth, debt payment problems, bankruptcy, and foreclosure from the Survey of Consumer Finances. Risk-taking behavior is assessed by stock market participation, by wider equity participation and by self-reported risk-taking in financial matters.

The findings indicate that financial distress is linked with lower participation in the stock market regardless of the financial resources and the demographic variables. But people's willingness to take financial risks does not change significantly, one way or another. This discrepancy between stated preference and actual behavior indicates that financial stress may lead to behavioural persistence through future changes to downside risk perception, and thus a decrease in the tendency to reengage with uncertain financial assets. The results make a unique contribution to the literature on household finance because they offer the first evidence that financial distress can have lasting behavioral consequences for the U.S. household sector.

2 Literature Review

Modern behavioural analysis of decision making under risk was based on the work of [1] Prospect Theory. Unlike the Expected Utility Theory, Prospect Theory suggests that utility is a function of the changes in one's final wealth, which it assumes. People consider results in terms of gains or losses based on a reference point. This shift in The framing effect is a fundamental change in how people evaluate a risky opportunity. The most important meaning of Prospect Theory is loss aversion: the disutility of losing a given amount is more beneficial than getting the same amount. The reflection effect[1] reveals that people tend to be loss averse and risk-seeking when they are making losses. This is an asymmetry that suggests that losses are more significant than gains and can influence future financial decisions.

The ramifications of financial conduct are significant. Because investment decisions are made with, Loss aversion may impact asset allocation and market participation in uncertain gains and losses. decisions. Households could also be deterred from making potentially accretive

investments due to the possibility of losing their investment. Losses are disproportionately greater compared to expected benefits. This behavioral mechanism The provision of insurance has become a key focus in the analysis of household financial behaviour in the presence of risk and uncertainty.

Financial economists have investigated the effects of loss aversion, based on Prospect Theory, in real-world markets. Hwang and Satchell [2] explore the intensity of loss aversion of financial market investors and conclude that loss aversion may be greater in the real financial markets than in the laboratory setting. Instead, their research suggests that real money can make behavioral reactions more intense, and that the loss aversion coefficient of $\tilde{2}.25$, is not the standard value. Most importantly, Hwang and Satchell (2010) also present evidence that loss aversion is different across market conditions. Their findings indicate that investor sensitivity to losses varies across bull and bear market cycles, implying that response to losses is not necessarily constant.

Yang expands upon the concept of loss aversion in financial markets in [3] by linking it to some longstanding financial puzzles. A puzzle is that many households do not invest in stock markets despite their high long-run returns. Vissing-Jørgensen (2002) reports significant variation in the composition of the household portfolio, and it reveals that there is still there considerable variation in the participation of households in the stock market when controlling for differences in wealth and income. This behavior is hypothesized by Yang [3] to be the result of loss aversion, whereby households may overvalue the downside risk compared to the upside potential. In addition, behavioral preferences are critical for the household portfolio choice as the authors of [4] have found direct evidence that households with more loss-averse preferences are less likely to buy equity. In the same way, the disposition effect is another phenomenon caused by loss aversion, in which investors sell profitable assets prematurely and keep on with the loss assets for too lengthy an interval. Together, these studies provide

evidence that loss aversion persists. An important framework in the understanding of the financial behaviour of the household and the market outcome.

Although Prospect Theory provides insights into how gains and losses are weighed, one question remains unanswered: Does financial distress in the past influence financial behavior in the future? Household financial choices can be path-dependent, and past financial experiences can influence future choices in times of uncertainty. Financial distress can lead to behavioral persistence through raising the risk of loss and lowering the motivation to re-enter into risky financial transactions.

In future investment choices, households facing debt delinquency, negative net worth, or bankruptcy might be less willing to engage in financial risk-taking, limiting their involvement in financial risk-taking activities. This mechanism follows the general pattern of the behavioral predictions of prospect theory, but it has not been as directly studied in the context of household finance.

Loss aversion is largely ignored in the literature, with its role traditionally viewed as a fixed behavioural constant, rather than an influence of financial distress in the real world on subsequent household financial behaviour. The paper seeks to fill this gap and to test whether financial distress indicators, which can be observed, are correlated with less participation in risky financial markets among U.S. households. The method is not experimental, and therefore does not provide data on financial behaviors in a controlled environment, but does reflect actual financial behavior in the real world within naturally occurring economic conditions. The analysis connects financial distress to observable investment choices, and yields new evidence of the impact of financial difficulties on future investment choices. household financial decision-making.

3 Data

The data analyzed in this study comes from the household component of the Survey of Consumer Finances (SCF), a nationally representative cross-sectional survey created by the Board of Governors of the Federal Reserve System [5]. The SCF is considered to be the most comprehensive and authoritative source of U.S. household balance sheet statistics. The SCF offers detailed information across a range of areas of household balance sheets, income, debt obligations, assets, and financial decision making behavior that is appropriate to studying the link between financial distress and the financial behavior of households. The analytical sample is composed of 22,975 households that have complete information for the variables of interest. The SCF has a complex survey design and empirical specifications include household sampling weights to ensure national representativeness.

The key outcome of interest is stock participation, a binary indicator (1 if stock ownership, 0 otherwise). This measure is the primary behavioral indicator of financial risk taking, since the presence of stocks is an actual exposure to unpredictable returns. Two robustness measures are used to test this, equity participation (which reflects general ownership of equity-based financial assets) and willingness to take financial risk (which is a self-reported financial risk tolerance measure). A key explanatory variable is a constructed variable which reflects prior-loss financial stress. Since the SCF does not directly ask about losses that occurred prior to the survey, indicators of negative net worth, late debt payments, bankruptcy within the last five years, and foreclosure within the last five years are used to proxy for financial distress. If one or more of these distressing conditions is experienced, the household is considered to be in prior loss.

To account for confounding factors associated with portfolio participation, the analysis includes a set of demographic and financial controls. Demographic controls include age, age

squared, education, marital status, number of children, labor force participation, and race. Financial controls include the natural logarithm of household income, the inverse hyperbolic sine transformation of net worth, and the natural logarithm of household debt. Table 1 presents summary statistics for the main variables. Approximately 29.1% of households report stock ownership, while 16.5% meet the criteria for the prior-loss proxy. Households classified under the prior-loss proxy exhibit substantially lower median income and median net worth, suggesting a meaningful distinction in financial conditions.

Table 1: Descriptive Statistics

Statistic	Value
Observations	22,975
Stock participation	29.1%
Equity participation	63.0%
Willing to take financial risk	5.3%
Prior-loss proxy	16.5%
Negative net worth	6.9%
Late payment	11.2%
Bankruptcy in past 5 years	1.1%
Foreclosure in past 5 years	0.4%
Mean income	\$1,592,855
Median income	\$94,039
Mean net worth	\$19,956,403
Median net worth	\$384,500
Mean age	54.47

4 Methodology

The empirical approach relies on logistic regression models which are weighted according to the survey. The empirical approach involves estimating a link between prior financial distress and household risk-taking behavior with the use of survey-weighted logistic regression models. Logistic regression is appropriate because the main dependent variable (stock ownership) is a binary variable.

The minimum specification requirements are provided by:

$$Pr(Y_i = 1) = F(\beta_0 + \beta_1 PriorLoss_i + \epsilon_i) \quad (1)$$

This indicates the stock ownership by household i (Y_i), the financial distress proxy ($PriorLoss_i$), and the logistic cumulative distribution function ($F(\cdot)$).

The preferred specification includes control variables for demographics and finances to reflect the fact that there is observable variation across households that should be included:

$$Pr(Y_i = 1) = F(\beta_0 + \beta_1 PriorLoss_i + X_i' \beta + Z_i' \gamma + \epsilon_i) \quad (2)$$

where X_i is the set of demographic characteristics and Z_i is the set of household conditions related to their financial situation. Demographic and financial controls are similar to the rest of the household finance literature, which argues that financial literacy, income and wealth are important factors in household financial decision-making [6].

Empirical approach uses the cross-sectional variation in the financial distress status of the household to estimate the relationship between the financial distress status in the past and the financial behavior of the household. The composite prior-loss proxy is broken down into its components to further investigate the mechanisms behind the prior-loss effect:

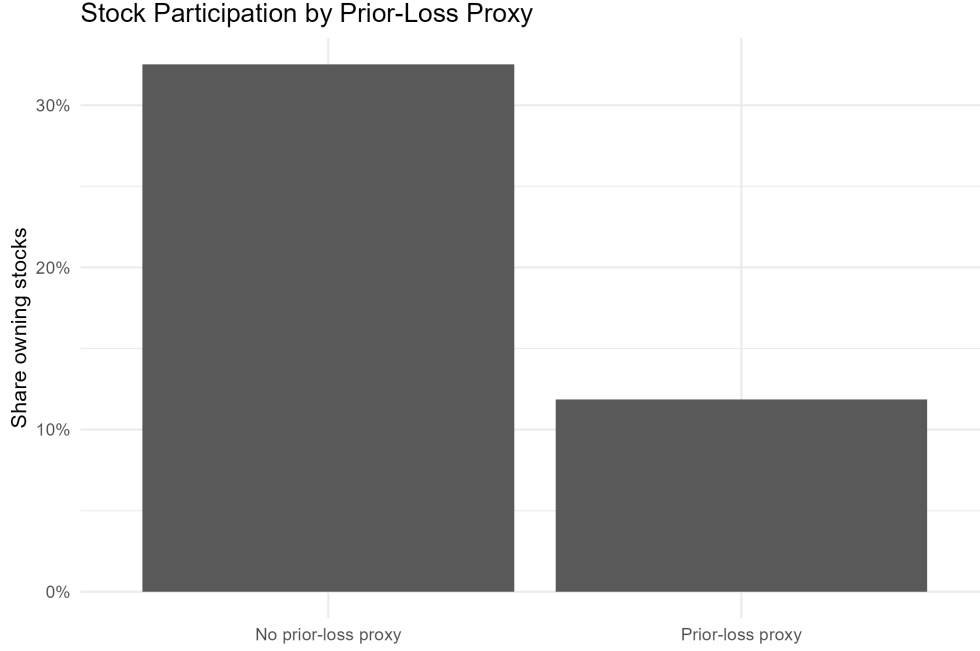
$$Pr(Y_i = 1) = F(\beta_0 + \beta_1 LatePayment_i + \beta_2 Bankruptcy_i + \beta_3 Foreclosure_i + X_i' \beta + Z_i' \gamma + \epsilon_i) \quad (3)$$

This breakdown allows for a recognition of different types of financial distress associations that are not necessarily homogeneous. All models are estimated with SCF survey weights

in order to make nationally representative inferences. Standard errors are not affected by heteroskedasticity. The empirical model is shaped by behavioral models of decision making under uncertainty, such as Prospect Theory [1], which posits the possibility of heightened sensitivity to downside risk due to prior losses. Instead of directly measuring parameters of loss aversion, this study investigates whether financial distress in the past represents a greater tendency to exhibit conservative revealed financial behavior. This approach would lead to a decrease in stock market participation by U.S. households in financial difficulty.

5 Results

The relationship between prior financial distress and stock market participation is shown in figure 1. The stock ownership rates are significantly higher for households with no prior-loss experience, with about 32% owning stocks compared to about 12% among households defined by the prior-loss proxy. This 20 percentage-point difference offers initial descriptive support for the main hypothesis that financial distress prior to the study is linked to a reduction in risky financial asset participation.



Source: Survey of Consumer Finances. Prior-loss proxy includes negative net worth, late payment, bankruptcy, or foreclosure.

Figure 1: Plans to participate in the stock market as a result of prior financial distress.

The main econometric results of the logistic regressions computed with the survey weights are reported in Table 2. The table in Appendix Table 5 shows the alternative outcome specifications.

Table 2: Main Econometric Results

Variable	Stock: Simple	Stock: Controls	Stock: Full	Distress Components
Prior-loss proxy	0.398*** (0.028)	0.490*** (0.036)	0.829* (0.068)	NA (0.059)
Late payment	NA	NA	NA	0.631***
Bankruptcy, past 5 years	NA	NA	NA	0.214***
Foreclosure, past 5 years	NA	NA	NA	1.022
Log income	NA	NA	1.676***	1.655***
IHS net worth	NA	NA	1.052***	1.055***
Log debt	NA	NA	1.018***	1.020***
Age	NA	0.973**	0.932***	0.933***
Age squared	NA	1.000**	1.001***	1.001***
Education category 2	NA	2.569***	1.839***	1.838***
Education category 3	NA	4.309***	2.960***	2.922***
Education category 4	NA	9.407***	4.689***	4.554***
Married category 2	NA	0.559***	0.949	0.955
Children	NA	0.936**	0.913***	0.917***
Labor force	NA	1.550***	1.225**	1.233**
Race category 2	NA	0.782***	0.868**	0.878**
Demographic controls	No	Yes	Yes	Yes
Financial controls	No	No	Yes	Yes
Observations	22,975	22,975	22,975	22,975

Notes: Entries are odds ratios from survey-weighted logistic regressions. Standard errors are reported in parentheses. The dependent variable is stock ownership. * p < 0.05, ** p < 0.01, *** p < 0.001.

The baseline specification (Column (1)) shows the odds ratio is 0.398 at the 1% level. This suggests that the financial difficulty is associated with about 60.2% less likely to own stocks

relative to no recent financial difficulty.

Demographic controls are introduced in Column (2) and the estimated odds ratio increases to 0.490, but continues to be very highly statistically significant. This would suggest that financially distressed households have about 51% less chance of owning a stock. The degree of the significance weakens, but the persistence of the significance suggests that it is an independent predictor of the lower participation in risky financial markets following financial distress.

The preferred specification includes financial controls for household income, net worth and debt in addition to the one listed above, and is reported in Column (3). Given this more challenging specification, the estimated odds ratio is 0.829. This corresponds to about 17.1% reduced likelihood of being a stockholder, given similar demographic, debt and income levels. Although the magnitude of the effect is reduced significantly after controlling for financial capacity, the continuity of the effect implies that financial distress has effects on behavior that go beyond just the weight of financial constraints.

The control variables develop the way they should. There is a positive correlation between household income and net worth and the stock market participation, which implies that the higher the financial resources, the higher the willingness to take investment risks. The coefficient on income is economically meaningful and statistically significant in all specifications and net worth is always statistically positive, and significant, of stock ownership. There is also a positive gradient by education, which is similar to the positive gradient by financial literacy and information benefits reported in previous studies.

To examine the mechanisms of the prior-loss effect on the aggregate, the prior-loss proxy is broken down into its three components: a proxy for late payment behaviour, a proxy for bankruptcy history and a proxy for foreclosure history, as shown in the table below in

column (4). Results reveal that there is meaningful financial distress type heterogeneity. An estimated odds ratio of 0.631 ($p < 0.001$); a value of near 36.9% lower odds of stock participation, is associated with late payment behavior. This effect is even larger for bankruptcy, as shown by the odds ratio of 0.214 ($p < 0.001$), which represents about 78.6% lower odds of owning a stock. In contrast, the number of foreclosures is statistically negligible, indicating that the financial distress events have different effects on financial risk-taking in the future.

The results suggest that the "aggregate impact of financial distress" observed is largely explained by financial liquidity stress and by extreme housing-related disruptions, rather than by housing-related shocks. This separation is useful economically because it indicates that financial hardship might be more important than the nature of the asset loss in terms of the person's need for immediate and severe financial loss.

Robustness checks included other specifications involving wider equity participation and self-reported willingness to take financial risk. Prior-loss proxy is not well correlated with either general equity ownership or reported financial risk tolerance. Interestingly, there has been no significant change in self-reported financial risk taking and substantial drop in actual stock market participation. This implies that financial risk preferences are stable among U.S. households and that financial distress may affect the behavior of their investments. The findings combined are consistent with behavioral theories of decision-making under uncertainty, specifically the Prospect Theory, which argue that financial behaviors that reveal preferences are more conservative the more stressed they are financially.

6 Conclusion

This study investigates if financial distress has an effect on individuals' financial actions and decisions in the United States. The results presented here, drawn from the Survey of

Consumer Finances, indicate that even when financial measures are controlled for, such as income, wealth, and demographic factors, households with a history of financial difficulty are less likely to be engaged in the stock market [7].

The results indicated that financial difficulty may have persistence effects of behavior beyond financial limitation [1]. The relationship between bankruptcy and a decrease in stock market participation is the strongest, and this may be due to the emotional and psychological effect on the person of more severe financial problems [8].

One of the key findings of this study is that of comparing stated preferences with actual behavior. Households facing financial difficulties do not indicate a lower risk appetite, but their investment decisions can be seen to become more cautious [1]. This implies financial hardship can have an impact on behavior that people are not fully aware of or reporting.

Policy implications of these results also exist. Increasing access to financial markets could be insufficient if households are reluctant to invest due to financial difficulties. Financial recovery and financial education programs can be useful to help restore confidence and re-enter investment markets over time for programs targeting financial recovery and financial education [6].

There are several limitations to this study. Note that the SCF is cross-sectional and can not be used to directly monitor changes in behavior over time. Furthermore, financial distress may not capture all aspects of personal experience and emotional reactions to financial strain.

Future research may take advantage of panel data to explore the effect of financial behavior after a distress event and whether financial behavior of households returns to risky financial markets over time. The psychological processes of these behavioural shifts could also be investigated in greater depth.

A Appendix

Table 3: Risk-Taking Outcomes by Prior-Loss Proxy

Prior-loss proxy	Observations	Stock participation	Equity participation	Willing to take financial risk	Median income	Median net worth
No prior-loss proxy	19,192	32.5%	68.1%	5.2%	\$113,496	\$591,600
Prior-loss proxy	3,783	11.8%	37.6%	6.0%	\$48,641	\$7,000

Table 4: Predicted Probability of Stock Ownership

Group	Predicted probability of stock ownership
No prior-loss proxy	28.7%
Prior-loss proxy	26.0%

Table 5: Alternative Outcome Models

	Equity Participation	Willing to Take Financial Risk
Prior-loss proxy	1.005 (0.071)	0.999 (0.114)
Log income	2.684*** (0.192)	0.992 (0.053)
IHS net worth	1.077*** (0.005)	0.995 (0.007)
Log debt	1.032*** (0.006)	1.006 (0.009)
Demographic controls	Yes	Yes
Financial controls	Yes	Yes
Observations	22,975	22,975

Notes: Entries are odds ratios from survey-weighted logistic regressions. Standard errors are reported in parentheses. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

References

- [1] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. [10.2307/1914185](#).
- [2] Soosung Hwang and Stephen E. Satchell. How loss averse are investors in financial markets? *Journal of Banking and Finance*, 34(10):2425–2438, 2010. [10.1016/j.jbankfin.2010.03.018](#).
- [3] Liyan Yang. Loss aversion in financial markets. *The Journal of Mechanism and Institution Design*, 4(1):119–137, 2019. [10.22574/jmid.2019.11.005](#).
- [4] Stephen G. Dimmock and Roy Kouwenberg. Loss-aversion and household portfolio choice. *Journal of Empirical Finance*, 17(3):441–459, 2010. ISSN 0927-5398. [10.1016/j.jempfin.2009.11.005](#).
- [5] Board of Governors of the Federal Reserve System. 2022 survey of consumer finances (scf): Summary extract public data. <https://www.federalreserve.gov/econres/scfindex.htm>, 2024. Summary variables used in the Federal Reserve Bulletin article; all dollar variables inflation-adjusted to 2022 dollars.
- [6] Annamaria Lusardi and Olivia S. Mitchell. The economic importance of financial literacy: Theory and evidence. *Journal of Economic Literature*, 52(1):5–44, 2014. [10.1257/jel.52.1.5](#).
- [7] Luigi Guiso, Paola Sapienza, and Luigi Zingales. Trusting the stock market. *Journal of Finance*, 63(6):2557–2600, 2008. [10.1111/j.1540-6261.2008.01408.x](#).
- [8] Atif Mian, Amir Sufi, and Francesco Trebbi. Foreclosures, house prices, and the real economy. *Journal of Finance*, 70(6):2587–2634, 2015. [10.1111/jof.12310](#).